

Q-P2FL: Quantum-Enhanced Federated Edge Intelligence for Privacy-Preserving Adversarial Attack Detection on Consumer Edge Devices

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Abstract—The rapid expansion of smart consumer environments is facilitated by device-to-device communication, which generates considerable quantities of data and improves the user experience. While this data provides useful insights, it is also subject to malicious cyberattacks. Machine Learning (ML)-based threat detection technologies solve these challenges; however crucial consumer electronics privacy concerns are often bypassed. The growing connectivity of consumer devices raises the possibility of adversarial cyberattacks. Quantum technology may give stronger protection against threats, while FL-based edge computing may help to ensure data privacy and security. This work proposes a Quantum-based Privacy-Preserving Federated Learning (Q-P2FL) technique for detecting adversarial attacks on consumer edge devices. First, as quantum computing advances, standard validation approaches may become obsolete. Quantum-based registration and authentication with Additive Homomorphic Encryption (AHE) is employed to safeguard privacy and secure model weights on edge devices. Second, image-based features are extracted from network traffic bytes to generate distinguished datasets. Third, adversarial examples are generated to assess the robustness of datasets. This is achieved by employing four distinct types of adversarial techniques that incorporate perturbations into the input data. The pre-trained Vision Transformer (ViT) extracts features to generate the local model weights. Finally, the Q-P2FL approach detects and classifies adversarial attacks, ensuring data security and privacy. The proposed method is evaluated on two standard datasets: Edge-IIoT and CICIoMT2024, with detection accuracies of 99.38% and 99.41%, respectively. This approach presents a promising solution for protecting consumer edge devices from adversarial attacks and improving their privacy-preserving capabilities.

Index Terms—Quantum computing, consumer devices, edge computing, federated learning, adversarial attacks, cybersecurity.

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I. INTRODUCTION

THE EXPANSION of consumer peripheral devices has significantly influenced the evolution of contemporary ecosystems, resulting in more personalized services and adaptive solutions. These devices, which range from smart home systems to wearable technology, generate and analyze vast quantities of private data, providing unique perspectives on individual preferences and actions. However, the interconnectivity of this ecosystem has posed substantial challenges to protecting data privacy and security. The integrity and dependability of these systems are at risk due to adversarial attacks, including ransomware, Denial-of-Service (DoS), and man-in-the-middle attacks. Consequently, robust defenses are necessary to maintain functionality and confidence [1]. Figure 1 demonstrates that traditional malware detection algorithms and antivirus software based on signatures frequently fail to address modern cyber threats. A potential alternative is the integration of ML into threat detection systems. These systems, however, are susceptible to privacy concerns because they rely on centralized infrastructures for model training. Sensitive data transported to centralized systems is vulnerable to breaches and unwanted access, underlining the need for more secure, decentralized decisions [2], [3].

Federated Learning (FL) has become an important step forward in addressing privacy concerns. FL ensures that data is maintained on local devices during the learning process by transferring only model changes, not unprocessed data. Furthermore, this methodology promotes privacy and allows for collaborative learning in distributed environments. However, FL is still vulnerable. Attackers can utilize techniques such as the Fast Gradient Sign Method (FGSM), Iterative Gradient Sign Method (IGSM), Projected Gradient Descent (PGD), or a hybrid approach to compromise the security and resilience of FL systems. These methods are referred to as evasion assaults and involve the use of adversarial samples to deceive the model into producing inaccurate predictions [4]. In our research, we assess the robustness of the classification task by employing these strategies to produce adversarial images. Furthermore, the current FL implementations use Fully Homomorphic Encryption (FHE) and Secure Multi-party Computation (SMC), which have large processing costs, limiting their usefulness in resource-constrained edge applications. Quantum technology improves the prevention of cyber threats, while edge computing enables data processing,

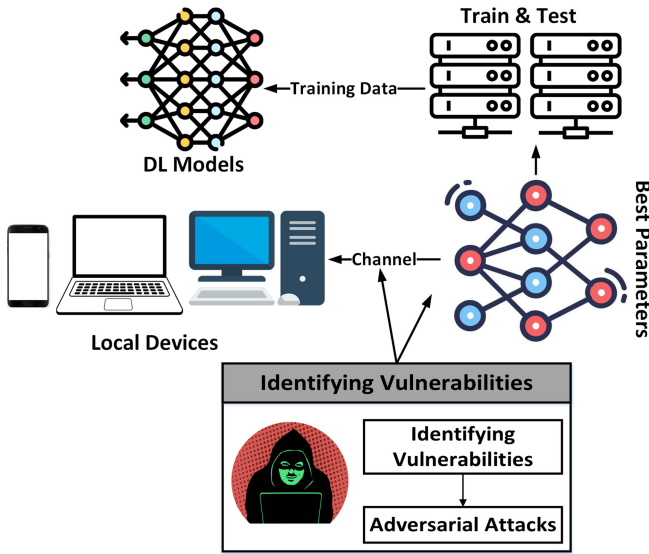


Fig. 1. Impact of adversarial attacks on traditional model training methods.

thereby reducing the vulnerabilities associated with centralized infrastructures [5], [6]. Quantum-based FL employs advanced quantum authentication to secure edge device communication while maintaining privacy. It employs quantum security techniques to defend against adversarial attacks. In contrast to conventional methods, it utilizes the Quantum Conference Key Agreement (QCKA) to establish high-security key exchange protocols. This provides quantum states impermeable in digital encryption, as any interception or duplication is immediately detectable, even under adverse conditions. Reliable interaction is ensured by this innovative technique, which safeguards critical data within decentralized FL systems [7], [8].

To address these difficulties, this study presents Q-P2FL, an FL framework for detecting adversarial attacks on consumer edge devices. The method protects against data breaches and model manipulation by utilizing quantum computing to overcome the limitations of conventional cryptography. Q-P2FL incorporates Additive Homomorphic Encryption (AHE) to balance data privacy with computational efficiency, making it beneficial for edge intelligence applications [9]. This method addresses consumer edge device heterogeneity and resource constraints by incorporating quantum-driven registration and authentication procedures to protect FL systems from adversarial threats [10], [11]. Q-P2FL enhances the security and scalability of federated edge intelligence systems, making consumer edge devices more resilient and privacy-preserving. The main contributions are given below:

- 1) This study leverages image-based features to effectively capture structured data categories, such as headers, procedures, and storage. The datasets used in this research are generated by systematically processing large volumes of network traffic packets. These customized datasets are then evaluated to develop a lightweight intrusion detection approach.
- 2) To reduce overfitting and class imbalance, data augmentation techniques are used to increase dataset diversity. Adversarial examples are produced utilizing FGSM, IGSM, PGD, and a hybrid IGSM-PGD method to

assess and strengthen resilience to evasion attacks. This approach enhances model performance, data security, and dataset resilience to adversarial threats.

- 3) We presented a novel Q-P2FL framework that combines quantum technology and FL to improve the security and privacy of consumer edge devices [12]. By employing quantum-based registration, authentication, and AHE, the architecture effectively protects model weights. This approach protects data privacy while also providing effective resistance against malicious attacks.
- 4) We conducted experiments using different numbers of edge devices to assess the performance of the proposed approach. Across all test scenarios, the method consistently achieved superior classification results, demonstrating its scalability and effectiveness.

The remaining sections of the paper are organized as follows: Section II discusses related work, and Section III describes the proposed methodology. In Section IV, the experimental results and discussions are presented, and the study is concluded in Section V.

II. RELATED WORK

Quantum-based FL approaches are gradually replacing conventional ML-based methods in the detection of adversarial intrusions in consumer electronics [2], [8], [12], [13]. The following section examines a variety of methods that utilize ML and quantum-based FL to identify and categorize attacks on consumer electronic devices.

A. ML/FL Methods

Qiu et al. [14] have created a novel adversarial attack that targets network intrusion detection systems based on the Internet of Things (IoT). This attack utilizes closed-box access to the Deep Learning (DL) model. Model extraction duplicates the closed-box model with minimum training data, and saliency maps highlight the most important packet features in detection results. These methods facilitate the efficient generation of adversarial examples by utilizing conventional models. Pawlicki et al. [15] used four previously proposed approaches to generate adversarial attacks that could damage a well-optimized intrusion detection algorithm during testing and proposed a mechanism to detect them. The four techniques for creating adversarial attacks and the appropriate context for artificial neural networks are described. Yamany et al. [4] used FL and network traffic adversarial contexts to improve medical device security with edge intelligence. An image-based dataset is generated by converting bytes of network traffic. The dataset is skewed due to the sporadic nature of client interactions, which may impair effectiveness. Four adversarial attack techniques are implemented to generate instances and evaluate the dataset's resilience. Ibitoye et al. [16] used the UNSW Canberra Cyber BoT-IoT dataset to evaluate Self-normalizing Neural Networks (SNNs) and Feed-forward Neural Networks (FNNs) in identifying breach vulnerabilities in an IoT network. The findings show that the FNN performs better than the SNN in terms of accuracy, precision, recall, and Cohen's

Kappa score for intrusion detection. However, the SNN performed better against adverse samples from the IoT dataset. Chen et al. [17] introduced the privacy score and evasion rate for FL-based intrusion detection, which quantifies traffic attribute similarity and adversarial attack impact. Experiments showed that hostile communications circumvented the state-of-the-art SOTA Kitsune technique, proving that existing protections are insufficient. They presented an optimization-based security strategy that reduces gradient distance while increasing input distance to improve FL-based intrusion detection and privacy.

B. Quantum-Based FL Methods

Namakshenas et al. [2] used quantum-based FL to address two significant challenges in consumer threat identification. The first issue addressed is the importance of rigorous customer validation in FL. The advent of quantum computing may render traditional validation methods useless. To address this issue, a quantum-centric enrollment and authorization approach is provided, which ensures thorough client evaluation inside an FL paradigm. The second challenge is to protect the model weights in FL. The incorporation of AHE ensures the privacy of FL users and efficient computation. Houda et al. [8] developed a privacy-preserving and efficient network intrusion detection paradigm based on quantum computing and FL. While safeguarding user privacy, decentralized FL enabled a multitude of consumer devices to train a global model. The efficiency of model training and inference is enhanced by quantum computation. Results showed significant improvements in processing efficiency and detection accuracy when compared to conventional approaches. Friha et al. [18] presented a differentially FL-based decentralized secure intrusion detection system for smart industrial locations. Quantum computing components include a key exchange protocol for peer-to-peer weight communication, a differentially private gradient exchange mechanism for FL privacy, and a decentralized FL framework. The decentralized framework reduces the possibility of aggregation server failure or attack in traditional FL systems. The approach identified several types of Industrial IoT cyberattacks in a real-world IIoT dataset. Yamany et al. [19] presented a quantum-based FL system capable of autonomously changing FL hyperparameters in autonomous vehicle settings during adversarial attacks. A quantum-behaved particle swarm optimization technique is employed to adjust the learning rate and local & global epochs. The proposed technique is integrated into a cyber-defense system to mitigate the impact of malicious attacks.

The proposed Q-P2FL approach improves the security and privacy of consumer devices in response to adversarial attacks. It integrates quantum-based registration, authentication, and AHE to enhance the privacy and protection of model weights. Additionally, image-based features retrieved from network traffic are used to generate novel datasets and tested for robustness using adversarial methods. The Q-P2FL approach efficiently identifies and classifies adversarial attacks, providing a reliable privacy-preserving solution for edge devices.

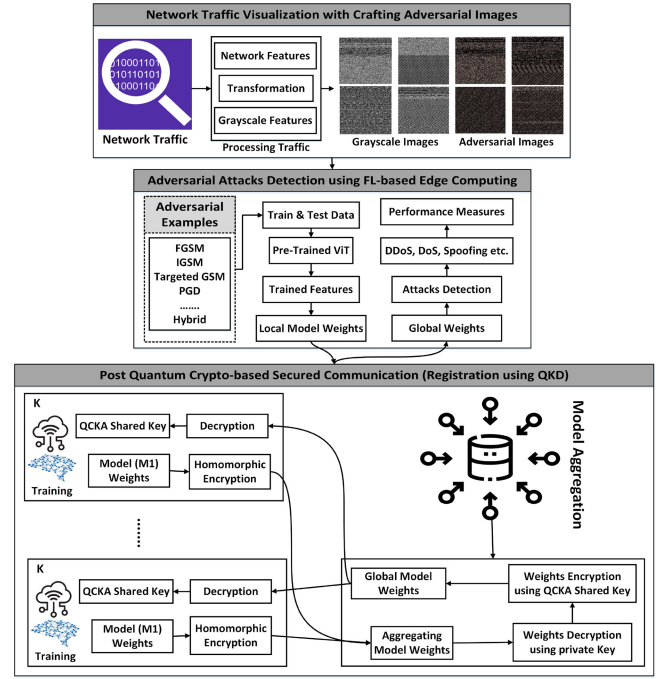


Fig. 2. Quantum-enhanced federated edge intelligence framework for detecting adversarial attacks on consumer edge devices.

III. PROPOSED METHODOLOGY: QUANTUM-BASED PRIVACY-PRESERVING FL

Figure 2 illustrates the proposed Q-P2FL framework. Quantum-based registration, which uses AHE, protects edge device privacy and model weights. Network traffic data is acquired using a customized crawling method and turned into images to generate adversarial examples. These are then classified using FL to ensure the system's resilience.

A. Dataset Preparation

The proposed approach is assessed using two widely recognized datasets. The first, Edge-IIoTset¹ [20], dataset is generated using an IoT/IIoT testbed that includes a range of devices, sensors, protocols, and edge configurations. It includes data from over ten IoT devices, including low-cost humidity and temperature sensors. It addresses 14 different forms of attacks, including malware, injection, man-in-the-middle, and DoS/DDoS. The data, which originally had 1176 attributes, has been reduced to 61 key features. This dataset provides a wide range of attack scenarios, making it useful for testing industrial and IoT intrusion detection systems. The second dataset primarily focuses on the security of healthcare devices, CICIoMT2024² [21]. It explores attacks on 40 devices (25 physical and 15 simulated) using protocols such as Bluetooth, MQTT, and Wi-Fi. The most prominent attacks are spoofing, reconnaissance, DDoS, and MQTT-based exploits. The security dataset is generated by replaying network traffic between switches and IoMT devices in real time using a network probe.

¹<https://www.kaggle.com/datasets/mohamedamineferrag/edgeiiotset-cyber-security-dataset-of-iiot-iiot/code>

²<https://www.unb.ca/cic/datasets/iomt-dataset-2024.html>



Fig. 3. Distribution of images in the CICIoMT2024 dataset.

B. Light-Weight Feature Extraction

Packet Capture (PCAP) files provide essential packet logs that enhance network performance and security. They help detect threats, identify unusual behavior, and track network issues during incident investigations. PCAP files are also critical for protocol analysis, aiding in the detection of errors and illegal activities [22], [23]. Packet data analysis can be used to measure performance more accurately, revealing latency issues and barriers. In addition to facilitating the identification of threats by exposing aberrant behaviors and potential security risks, PCAP files serve as an audit trail, ensuring adherence to laws and security policies. They help troubleshoot network faults by revealing individual data flows. Addressing static and dynamic classification problems is becoming increasingly important with the growth of network traffic. We explored using visual signals to detect malicious communications as a solution. This method employs image processing to convert network traffic patterns into textural features, eliminating the necessity for traffic signatures or reverse engineering. PCAP file analysis uses byte sequences that show harmful changes to transform 8-bit vectors into grayscale images. A structured image representation of the network traffic is formed by mapping each byte in the sequence to a pixel intensity value. This transformation allows for the visibility of patterns associated with harmful activity, resulting in more efficient intrusion detection. Subsequently, the dimensions of these images are reduced to 128×128 pixels. Image-based characteristics are faster and utilize fewer resources than previous approaches due to their small size. They also uncover subtle patterns that conventional approaches may overlook, making them useful for identifying malicious attacks [24]. Figure 3 shows the number of extracted images using the CICIoMT2024 train dataset. The lightweight nature of these capabilities makes it easier to detect possible vulnerabilities in real time and respond quickly to security threats.

C. Synthetic Data Generation

Synthetic data generation is an effective strategy for increasing feature extraction and addressing class imbalances in

network traffic images, especially when identifying malicious behavior. By producing synthetic data, the model can gain knowledge from a greater variety of network patterns, improving its ability to identify anomalies [25], [26].

To simulate different orientations of the image, a synthetic rotation by an angle θ can be described in equation (1):

$$I_{\text{synthetic}} = \mathcal{R}_{\theta}(I) \quad (1)$$

where $\mathcal{R}_{\theta}$ denotes the operation that rotates image I by angle θ .

Scaling the image to new dimensions $x \times y$ is another important transformation in synthetic data generation shown in equation (2):

$$I_{\text{scaled}} = I \cdot \mathcal{S}(x, y) \quad (2)$$

where $\mathcal{S}(x, y)$ is the scaling function that resizes the image I to the dimensions $x \times y$.

Incorporating noise into the image for enhanced robustness can be modeled as shown in equation (3):

$$I_{\text{noisy}} = I + \epsilon, \quad \epsilon \sim \mathcal{N}(\mu, \sigma) \quad (3)$$

where ϵ is noise sampled from a Gaussian distribution with mean μ and standard deviation σ .

For simulating variations in data composition as shown in equation (4), synthetic cropping is applied to the image, adjusting its size to $x' \times y'$:

$$I_{\text{cropped}} = I[x' \times y'] \quad (4)$$

where $x' \times y'$ indicates the target dimensions of the cropped image.

Synthetic color jittering, which modifies the image's brightness, contrast, saturation, and hue, is represented in equation (5):

$$I_{\text{jittered}} = I \odot \mathcal{J}(\delta_{\text{brightness}}, \delta_{\text{contrast}}, \delta_{\text{saturation}}, \delta_{\text{hue}}) \quad (5)$$

where $\mathcal{J}(\delta_{\text{brightness}}, \delta_{\text{contrast}}, \delta_{\text{saturation}}, \delta_{\text{hue}})$ applies random changes to these properties.

Horizontal and vertical flipping, key for generating different image perspectives, can be formalized in equation (6):

$$I_{\text{flipped}} = \begin{cases} \mathcal{F}_h(I) & \text{with probability } p_{\text{horizontal}} \\ \mathcal{F}_v(I) & \text{with probability } p_{\text{vertical}} \end{cases} \quad (6)$$

where $\mathcal{F}_h(I)$ and $\mathcal{F}_v(I)$ represent the horizontal and vertical flip operations.

Lastly, applying an affine transformation to the image for synthetic data generation can be written in equation (7):

$$I_{\text{affine}} = \mathcal{A}(I, \mathbf{M}, \mathbf{b}) \quad (7)$$

where $\mathcal{A}(I, \mathbf{M}, \mathbf{b})$ applies the affine transformation using matrix $\mathbf{M}$ and translation vector $\mathbf{b}$.

These strategies are used to produce synthetic data, which makes the training dataset more diverse and allows the model to learn more robust and generalized features. This enables the model to more effectively identify malicious traffic, even in scenarios where the network conditions are dynamic or the data is insufficient. Additionally, synthetic data production ensures that the model is trained on a wide range of real-world traffic patterns and anomalies. For intrusion detection

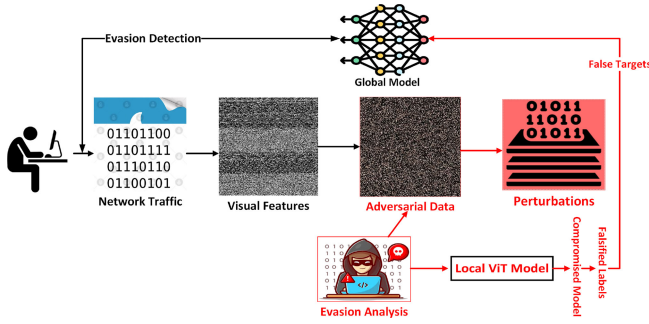


Fig. 4. Examining evasion attacks based on irregularities in network traffic flows.

applications, this method enhances both generalization and accuracy.

D. Synthesizing Adversarial Perturbations

Evasion attacks are a common adversarial strategy in which attackers intentionally alter inputs to confuse ML models. These attacks take advantage of weaknesses in the model by making small changes to legitimate inputs, causing them to be misclassified as shown in Figure 4. This study evaluates four adversarial techniques: PGD, FGSM, IGSM, and a hybrid approach that combines PGD and IGSM [27]. These strategies are used to evaluate their efficacy in creating adversarial examples that can fool the target model while also lowering the degree of perturbation introduced. Understanding and protecting against evasion attacks requires the ability to synthesize suspicious perturbations. The examination of the robustness of intrusion detection against evasion attacks offers information about their ability to effectively manage sophisticated cyber intrusions [28].

By utilizing adversarial perturbation approaches, it is possible to predict evasion risks and improve the adaptability of intrusion detection. In addition to revealing potential vulnerabilities during model testing in these scenarios, countermeasures are also implemented to mitigate these attacks. By predicting and resolving evasion strategies, it is possible to detect Advanced Persistent Threats (APTs), malware, and other cyberattacks that employ minor modifications to evade detection [14], [29].

1) *FGSM*: FGSM uses a gradient-based, one-step technique for adversarial sample generation. It changes the input $\mathbf{x}$ in the direction of the gradient of the loss function $\mathcal{L}(\mathbf{w}, \mathbf{x}, \mathbf{y})$, where $\mathbf{w}$ represents the model coefficients and $\mathbf{y}$ represents the true label. Equation (8) calculates the adversarial input $\mathbf{x}'$, whereas λ determines the perturbation magnitude.

$$\mathbf{x}' = \mathbf{x} + \lambda \cdot \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{w}, \mathbf{x}, \mathbf{y})). \quad (8)$$

This method is computationally efficient, making it appropriate for fast adversarial attacks. However, its simplicity limits its effectiveness against strong defensive strategies.

2) *IGSM*: The FGSM concept is employed iteratively by IGSM, which is an acronym for Basic Iterative Method (BIM). This technique uses smaller perturbation steps δ across multiple rounds to ensure that adversarial perturbations accumulate gradually without exceeding a predetermined limit.

Equations (9) and (10) specify the iterations, starting from the initial input $\mathbf{x}$:

$$\mathbf{x}'^{(0)} = \mathbf{x}, \quad (9)$$

$$\mathbf{x}'^{(t+1)} = \text{clip}_{\mathbf{x}, \lambda} \left(\mathbf{x}'^{(t)} + \delta \cdot \text{sign} \left(\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{w}, \mathbf{x}'^{(t)}, \mathbf{y}) \right) \right), \quad (10)$$

where t is the iteration step, and $\text{clip}_{\mathbf{x}, \lambda}(\cdot)$ constrains the adversarial input to the perturbation range described by λ . This repeated technique improves the relevancy of adversarial examples.

3) *PGD*: PGD improves IGSM by including a projection operation that keeps perturbations under a L_p -norm limit. It uses the initial input as a starting point for each iteration and applies perturbations to return the adversarial example to a suitable range. For example, the formulation is indicated in equations (11) (12).

$$\mathbf{x}'^{(0)} = \mathbf{x}, \quad (11)$$

$$\mathbf{x}'^{(t+1)} = \Pi_{\mathcal{B}(\mathbf{x}, \lambda)} \left(\mathbf{x}'^{(t)} + \delta \cdot \text{sign} \left(\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{w}, \mathbf{x}'^{(t)}, \mathbf{y}) \right) \right), \quad (12)$$

The projection operator on the L_p -norm ball of radius λ around the original input $\mathbf{x}$ is represented by $\Pi_{\mathcal{B}(\mathbf{x}, \lambda)}(\cdot)$. This ensures that the generated adversarial examples adhere to the specified perturbation limitations.

4) *Hybrid Method (PGD and IGSM)*: The hybrid technique combines the optimal features of both IGSM and PGD, combining their iterative nature with resilient optimization. It initiates the process by generating adversarial samples using IGSM, which are subsequently refined using PGD. This two-step technique improves attack efficiency while preserving imperceptible disruptions, as illustrated in equations (13) (14) (15).

IGSM Phase:

$$\mathbf{x}'^{(0)} = \mathbf{x}, \quad (13)$$

$$\mathbf{x}'^{(t+1)} = \text{clip}_{\mathbf{x}, \lambda} \left(\mathbf{x}'^{(t)} + \delta \cdot \text{sign} \left(\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{w}, \mathbf{x}'^{(t)}, \mathbf{y}) \right) \right). \quad (14)$$

PGD Refinement Phase:

$$\mathbf{x}'^{(t+1)} = \Pi_{\mathcal{B}(\mathbf{x}, \lambda)} \left(\mathbf{x}'^{(t)} + \delta \cdot \text{sign} \left(\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{w}, \mathbf{x}'^{(t)}, \mathbf{y}) \right) \right). \quad (15)$$

This hybrid approach emphasizes the benefits of both methods, with IGSM enabling precise initialization. In contrast, PGD ensures that adversarial instances adhere to perturbation requirements, resulting in more effective and resilient samples. These developments increase the detection and analysis of sophisticated cyberattacks, including evasion attacks, allowing for proactive countermeasures in the real world. The pre-trained ViT, which is renowned for its exceptional performance in image processing and feature extraction, is utilized to extract features for local model training [30], [31], [32]. The ViT architecture efficiently captures complex patterns and dependencies inside network traffic data by using self-attention methods. In this work, ViT is applied on the client side to process adversarial instances produced from network traffic, therefore improving its capacity to identify abnormalities, including DDoS attacks. The feature map extracted by the ViT model during training is illustrated in Figure 5, which demonstrates its ability to learn unique representations. The ViT model updates model weights depending on processed adversarial examples by local training on the client device.

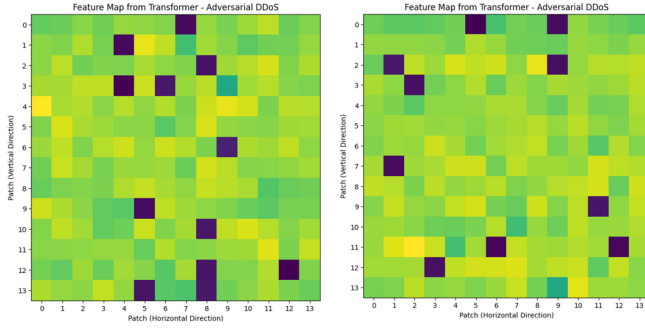


Fig. 5. Cyberattack features map utilizing pre-trained vision transformer.



Fig. 6. Training times distributed across edge devices.

Figure 6 shows the distributed training time among separate nodes. After local training, the model weights are sent to the server for global aggregation, enabling collaborative learning and enhancing detection performance.

E. Quantum-Based Secured FL

The proposed Q-P2FL design effectively protects user privacy and detects adversarial attacks on edge devices by utilizing quantum-based cryptography and AHE.

F. Quantum-Based Cryptography for Privacy Preservation

Quantum Key Distribution (QKD) is used extensively to ensure a secure connection between the aggregation server and clients, as shown in Algorithm 1. During QKD, quantum channels are utilized for exchanging cryptographic keys. An alarm would be triggered regarding potential security vulnerabilities in the system if an attempt to eavesdrop on the key exchange disrupts the quantum state [6], [19]. To encrypt and authenticate communications, both the client and the aggregation server use shared keys (k_i).

Client-aggregation server mutual authentication is secured via the Quantum Classical Key Agreement (QCKA) protocol as shown in equation (16):

$$k_{\text{shared}} = \text{QCKA}(k_i, k_j) \quad (16)$$

The term k_j refers to one of the keys involved in the Quantum Classical Key Agreement (QCKA) protocol. Specifically, k_j represents the key associated with the second party (typically the server or another client) in the key exchange process. In the QCKA protocol, two parties (such as a client and a server) exchange keys, k_i (the client's key) and k_j (the server's key or another client's key). These keys are subsequently employed to generate a shared secret key,

Algorithm 1: Quantum-Based Secured FL for Adversarial Attack Detection

Input: Client datasets: D_i for $i = 1, 2, \dots, N$

Output: Global model for adversarial attack detection

$\mathcal{M}_g$

Function QuantumSecuredFL($D_1, D_2, \dots, D_N$):

Initialize client datasets D_i for $i = 1, 2, \dots, N$;

Initialize QKD keys for secure communication: k_i for each client i ;

Initialize quantum classical key agreement for client-server mutual authentication:

$k_{\text{shared}} = \text{QCKA}(k_i, k_j)$;

for each client i do

Train local model: $w_i \leftarrow \text{Train}(D_i)$ // Train local model on client data D_i ;

$E(\theta_i) \leftarrow \text{Encrypt}(w_i)$ // Encrypt model weights for privacy preservation using AHE;

for each client i do

Send encrypted model update to server: Send $E(\theta_i)$ to aggregation server for global model aggregation;

Aggregation step: $E(\mathcal{M}_g) = \text{AHE}\left(\sum_{i=1}^N E(\theta_i)\right)$;

$w_{\text{global}}^{\text{encrypted}} \leftarrow \text{Aggregate}(E(\mathcal{M}_g))$ // Aggregate encrypted model updates;

Decryption step:

$w_{\text{global}} \leftarrow \text{Decrypt}(w_{\text{global}}^{\text{encrypted}}, k_{\text{shared}})$ // Decrypt aggregated model;

for each input x in test dataset do

$\hat{y}_i \leftarrow f(\mathcal{M}_g, x)$ // Use global model to predict and detect adversarial attack;

 Compare $\hat{y}_i$ with intrusion patterns to detect malicious activities;

Final Output: Return the global model $\mathcal{M}_g$ for adversarial attack detection;

k_{shared} . This shared key secures client-server communication, preventing attackers from impersonating valid clients or inserting malicious data into FL.

G. Additive Homomorphic Encryption for Privacy-Preserving Aggregation

AHE is used to ensure the privacy of model updates during the FL aggregation phase. HE allows computations on encrypted data to be performed without the need for decryption. As a result, the aggregation server can generate the global model weights $\mathcal{M}_g$ from encrypted updates without accessing the sensitive data underlying.

Let $E(\theta_i)$ represent the encrypted model weights for each client i as given in equation (17):

$$E(\mathcal{M}_g) = \text{AHE}\left(\sum_{i=1}^N E(\theta_i)\right) \quad (17)$$

This guarantees that the server may aggregate model updates from multiple clients without disclosing any sensitive information regarding the data or local models of any of the clients. HE encryption ensures that sensitive data remains confidential during the FL process, providing further protection against malicious attempts and data leaks.

H. Federated Learning for Adversarial Detection

The proposed approach uses federated averaging (FedAvg) to aggregate the global server model. This method guarantees the reliable and effective integration of locally trained models by computing the weighted average of model updates from client devices. Consequently, it ensures the privacy of data throughout the aggregation process. Clients in this system train local models to detect adversarial behaviors by minimizing a local loss function $\mathcal{L}_i$ as given in equation (18):

$$\mathcal{L}_i = \sum_{x \in D_i} \mathcal{L}(\hat{y}_i, y) \quad (18)$$

Afterward, QKD is used for secure communication, and encrypted model updates are merged into a global model $\mathcal{M}_g$ safely and privately. The global approach benefits from the diverse and secure contributions made by several clients, which improves the detection of malicious attacks.

I. Adversarial Attack Detection

The global model $\mathcal{M}_g$ is used to identify adversarial attacks by comparing its predictions to existing intrusion patterns. The detection function $\hat{y}_i$ is determined for a given input x using the following computation in equation (19):

$$\hat{y}_i = f(\mathcal{M}_i, x) \quad (19)$$

The prediction results for the i -th input, denoted as $\hat{y}_i$, are obtained by applying the model function f to the model $\mathcal{M}_i$ and the input data x , where $\hat{y}_i$ stores the complete output of this prediction. By incorporating the knowledge of all participating clients, the global model enhances its ability to identify adversarial activities on edge devices and increases its resilience to a variety of attack strategies.

The use of quantum-based cryptography provides an extra layer of protection, making the system resilient to eavesdropping and adversarial attacks. The proposed method improves adversarial attack detection by combining QKD, QCKA, and AHE, ensuring secure communication and privacy as shown in Algorithm 1. Clients use their datasets to train local models and then send encrypted updates to the server after encrypting the model weights with AHE. The server uses the common QCKA key to decrypt and aggregate the local models received from various clients. After comparing predictions to observed intrusion patterns, the final model is used to detect adversarial attacks. This approach ensures secure data transmission, privacy-preserving FL, and effective adversarial threat detection.

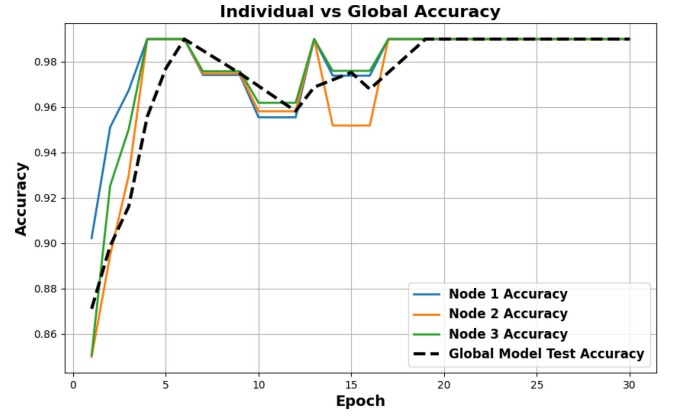


Fig. 7. Individual vs global accuracy epoch curves for 3 edge nodes.

IV. RESULTS AND DISCUSSIONS

A. Performance Indicators

We selected various network traffic-based images and several edge nodes to evaluate the proposed method comprehensively. The proposed method is evaluated using two publicly available standard datasets, Edge-IIoT, and CICIoMT2024. We measured performance using precision, recall, FL-score, and accuracy. These metrics are calculated from True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) counts. The performance measures are provided by Equations (20)-(23).

$$\text{Precision} = \frac{TP}{(TP + FP)} \quad (20)$$

$$\text{Recall} = \frac{FP}{(FP + TN)} \quad (21)$$

$$\text{FL-score} = \frac{(2 * TP)}{(2TP + FP + FN)} \quad (22)$$

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (23)$$

B. Performance Analysis and Comparisons

The proposed approach is tested across a variety of edge node variants to demonstrate its effectiveness. Figure 7 shows the dynamic epoch curves for each node and the global server with three edge nodes. Nodes 2 and 3 start at 82%, the node 1 at 91%, and the global server at 86%. The curves have an increasing trend and 99% accuracy up to the fifth epoch, after which they decrease to 95% and then grow again. The global curve exhibits the same continuous and normal behavior as the epoch curves of the other nodes. Figure 8 shows the dynamic loss values for three edge nodes. These values are behaving oppositely, as illustrated in Figure 7, which illustrates the consistent efficacy of the proposed approach. Figure 9 shows the dynamic accuracy curves for five edge nodes. The model may have initially struggled to identify the optimal parameters when it examined various parts of the loss, which could explain the large decline in accuracy between epochs 10 and 20. However, as training progresses and the model stabilizes, it begins to generalize more well, resulting in the observed increase in accuracy, which eventually reaches

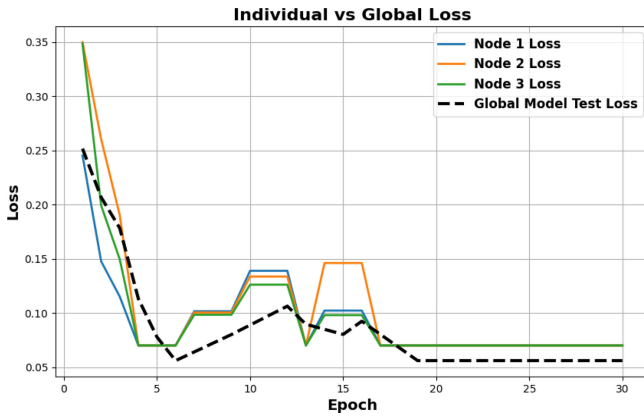


Fig. 8. Individual vs global loss epoch curves for 3 edge nodes.

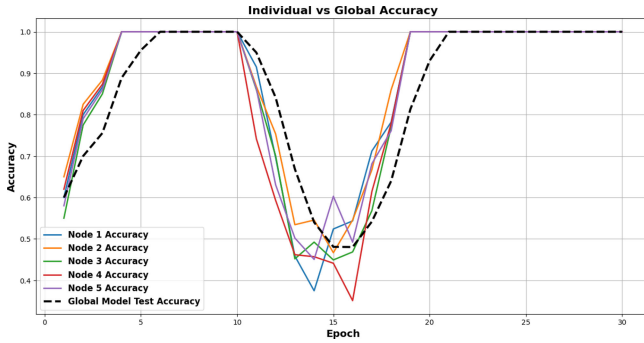


Fig. 9. Individual vs global accuracy epoch curves for 5 edge nodes.

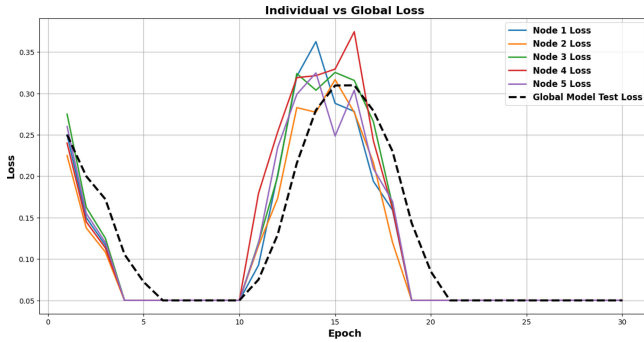


Fig. 10. Individual vs global loss epoch curves for 5 edge nodes.

99.60%. Similarly, Figure 10 depicts the loss values for 5 edge nodes that behave inversely to the values shown in Figure 9.

Table I compares the performance of adversarial attack detection using the EdgeIoT dataset. Four adversarial approaches are used: FGSM, IGSM, PGD, and a hybrid of IGSM and PGD. The precision, recall, FL-score, and accuracy performance indicators are utilized to assess the effectiveness of the proposed approach. The FGSM has the lowest performance because of its straightforward properties. The hybrid method provides its highest performance, with precision, recall, FL-score, and accuracy of 99.82%, 99.64%, and 99.15% for the benign class, and 99.06%, 99.46%, and 99.28% for the attack class, respectively. Overall, the accuracy is 99.38%. Table II compares the performance of the same indicators using the CICIoMT2024 dataset. It can

TABLE I
PERFORMANCE COMPARISON FOR ADVERSARIAL ATTACK DETECTION USING EDGEIoT DATASET

Adversarial	Traffic	Precision	Recall	FL-score	Accuracy
FGSM	Benign	93.14	97.46	95.62	94.39
	Attack	95.73	91.32	93.48	
IGSM	Benign	94.76	99.03	96.61	95.62
	Attack	98.51	90.68	94.62	
PGD	Benign	97.24	98.28	97.71	97.71
	Attack	97.89	97.84	97.66	
Hybrid	Benign	99.82	99.64	99.15	99.38
	Attack	99.06	99.46	99.28	

TABLE II
PERFORMANCE COMPARISON FOR ADVERSARIAL ATTACK DETECTION USING CICIoMT2024 DATASET

Adversarial	Traffic	Precision	Recall	FL-score	Accuracy
FGSM	Benign	96.24	91.67	93.9	94.28
	Attack	92.34	96.75	94.49	
IGSM	Benign	98.35	94.19	96.23	96.47
	Attack	94.57	98.66	96.57	
PGD	Benign	100	96.65	97.08	98.26
	Attack	97.52	100	98.65	
Hybrid	Benign	99.94	99.86	99.26	99.41
	Attack	99.29	99.82	98.93	

be seen that the hybrid approach is stronger and has higher performance metrics than individual methods such as FGSM, IGSM, and PGD.

Table III demonstrates the results of comparing the PGD and Hybrid techniques to classify adversarial attacks on the EdgeIoT dataset. We have shown only these two techniques as they deliver the highest classification results. The PGD technique performs somewhat worse for MITM and SQL Injection attacks, but it achieves good precision and recall for DDoS UDP Flood (98.89% and 98.77%) and Vulnerability Scanner (99.12% and 99.34%). In comparison, the Hybrid approach regularly beats PGD, with an accuracy of more than 96% for the majority of attack types. The results include a 100% recall for DDoS UDP Flood and a strong FL-score for Ransomware (98.88%) and XSS (99.32%). The hybrid model effectively detects adversarial attacks, despite significantly weaker performance on MITM and Password attacks. Table IV compares FGSM, IGSM, PGD, and Hybrid approaches for DDoS, DoS, MQTT, Recon, and Spoofing adversarial attack classification using the CICIoMT2024 dataset. FGSM is quite effective for spoofing (98.87% precision, 99.24% FL-score) and recon (96.09% FL-score), but less effective for DDoS and MQTT. IGSM maintains good performance for recon and spoofing while improving on MQTT (94.1% FL-score), however, it performs moderately for DDoS. PGD exhibits exceptional performance, particularly in the areas of DoS (96.96% FL-score) and Spoofing (99.77% precision). The Hybrid approach outperforms all others, receiving the highest scores across most attack types, including notable findings for Spoofing (99.52% FL-score) and MQTT (97.89% FL-score), demonstrating greater reliability in adversarial attack detection.

Table V displays the classification accuracy of Edge-IIoTset and CICIoMT 2024 datasets with varying edge nodes. The findings suggest that adding edge nodes consistently increases accuracy. In the Edge-IIoTset, accuracy rises from 92.46%

TABLE III
PERFORMANCE COMPARISON FOR ADVERSARIAL ATTACK CLASSIFICATION USING EDGEIIOT DATASET

Adversarial	Attack	Precision	Recall	FL-score	Accuracy
PGD	Backdoor	87.91	85.24	86.41	93.11
	DDoS HTTP Flood	88.91	91.34	90.82	
	DDoS ICMP Flood	91.62	92.01	93.11	
	DDoS TCP SYN Flood	93.84	85.78	90.44	
	DDoS UDP Flood	98.89	98.77	98.83	
	MITM (ARP spoofing + DNS)	87.89	88.12	82.94	
	OS Fingerprinting	99.54	99.12	98.01	
	Password	91.15	88.23	90.91	
	Port Scanning	98.94	95.64	97.28	
	Ransomware	97.85	99.72	98.77	
	SQL injection	85.48	88.63	87.98	
	Uploading	90.89	89.12	89.98	
	Vulnerability scanner	99.12	99.34	99.23	
	XSS	98.42	97.86	98.62	
Hybrid	Backdoor	88.57	95.67	91.95	96.04
	DDoS HTTP Flood	96.12	96.78	96.45	
	DDoS ICMP Flood	97.45	97.76	97.61	
	DDoS TCP SYN Flood	92.34	93.67	93.24	
	DDoS UDP Flood	99.82	100	99.66	
	MITM (ARP spoofing + DNS)	83.67	94.91	88.68	
	OS Fingerprinting	97.87	99.38	98.93	
	Password	95.32	79.12	87.11	
	Port Scanning	99.48	96.89	97.4	
	Ransomware	97.85	99.91	98.88	
	SQL injection	99.21	96.78	97.99	
	Uploading	97.56	99.91	98.28	
	Vulnerability scanner	99.87	98.52	98.76	
	XSS	99.45	99.78	99.32	

TABLE IV
PERFORMANCE COMPARISON FOR ADVERSARIAL ATTACK CLASSIFICATION USING CICIOMT2024 DATASET

Adversarial	Class	Precision	Recall	FL-score	Accuracy
FGSM	DDoS	91.32	93.56	90.11	89.86
	DoS	94.45	92.13	93.92	
	MQTT	90.67	90.23	92.11	
	Recon	99.12	95.05	96.09	
	Spoofing	98.87	97.92	99.24	
IGSM	DDoS	90.48	91.11	89.77	93.74
	DoS	91.36	92.91	91.63	
	MQTT	93.45	94.76	94.1	
	Recon	95.12	93.45	95.12	
	Spoofing	97.56	95.93	95.32	
PGD	DDoS	91.22	88.93	91.01	96.62
	DoS	96.26	95.89	96.96	
	MQTT	94.23	97.76	95.94	
	Recon	92.36	93.92	94.44	
	Spoofing	99.77	98.64	98.91	
Hybrid	DDoS	96.03	95.34	97.18	97.31
	DoS	97.63	98.21	97.91	
	MQTT	95.87	98.99	97.89	
	Recon	97.75	95.88	97.66	
	Spoofing	99.46	99.21	99.52	

TABLE V
COMPARISON OF CLASSIFICATION ACCURACY USING VARIOUS NUMBERS OF EDGE NODES

Datasets	Nodes	Accuracy
Edge-IIoTset	2	92.46
	5	94.03
	7	96.24
	12	99.38
CICIOMT 2024	2	90.81
	5	93.86
	7	97.22
	12	99.41

with two nodes to 99.38% with twelve nodes. Similarly, CICIOMT 2024 increases from 90.81% with 2 nodes to 99.41% with 12 nodes. FL performs better with more clients because it provides a more diverse and representative dataset, improving generalization. It also improves the model's robustness by reducing overfitting to individual client data. Furthermore, parallel client training speeds up learning, improving the model. Figure 11 shows the confusion matrix for intrusion detection using an EdgeIIoT dataset. The attack class has 99.2% detection accuracy, while benign has 99.6%. Figure 12 shows the confusion matrix for intrusion detection using the CICIOMT2024 dataset. The attack is classified as 99.3%, while the benign is classified as 99.9%.

Table VI shows the performance comparison of the proposed work with the related published works. Ullah et al. [24] used CNN-based FL for intrusion detection on edge devices. Further, it addressed the issue of intermittent client behavior and class imbalance. The proposed approach is tested on two datasets and provides 99.28% classification accuracy. Idrissi et al. [33] introduced Fed-ANIDS, a network intrusion detection system that uses anomaly detection and FL to overcome privacy problems in centralized architectures. The Fed-ANIDS model achieved 94.48% classification accuracy across prominent datasets such as USTC-TFC2016, CIC-IDS2017, and CSE-CIC-IDS2018. Thein et al. [34] introduced Personalized Federated Learning for Intrusion Detection Systems (pFL-IDS), a resilient method that integrates a poisoned client detector to reduce poisoning attacks while tackling the diversity of IoT data. The proposed approach provides 99.04% accuracy. Chen et al. [35] created VAN-IDS, a hybrid model that uses Bi-LSTM for packet analysis and LGBM for vehicle attributes to detect DoS and Sybil attacks with 98.5% accuracy. FL-based VAN-FED-IDS aggregates collaborative detection models using cloud services and

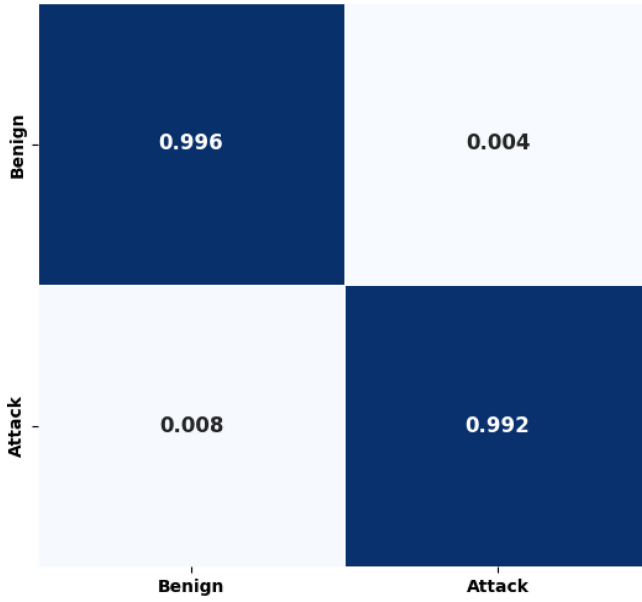


Fig. 11. Confusion Matrix using EdgeIoT dataset.

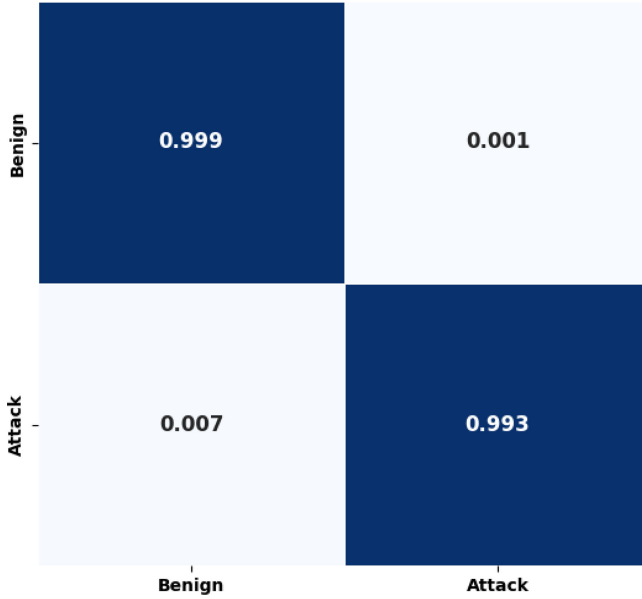


Fig. 12. Confusion Matrix using CICIoMT2024 dataset.

TABLE VI
PERFORMANCE COMPARISON WITH RELATED PUBLISHED WORKS

Work	Method	Accuracy
Ullah et al. [24]	CNN-based FL	99.28
Idrissi et al [33]	Fed-ANIDS	94.48
Thein et al. [34]	pFL-IDS	99.04
Chen et al. [35]	VAN-FED-IDS	98.5
Proposed Approach	Q-P2FL FL	99.41

manages local training through roadside units to improve privacy and training speed.

V. CONCLUSION

This study presents a Q-P2FL technique to mitigate the increasing threat of adversarial attacks on consumer

edge devices while ensuring adequate security and privacy. Combining AHE with quantum-based registration and authentication protects user privacy and model weights. Four perturbation techniques are employed to generate adversarial instances to test the resilience of the dataset, and the image-based properties of network traffic bytes are utilized for developing unique datasets. We used a pre-trained ViT to extract features and generate local model weights to detect adversarial attacks. The proposed method achieved 99.38% and 99.41% detection accuracies on the Edge-IIoT and CICIoMT2024 datasets, demonstrating its potential as a privacy-preserving solution for protecting consumer edge devices in smart environments. Despite its excellent detection accuracy, the proposed method may need optimization for large-scale, heterogeneous edge scenarios. The computational costs of quantum authentication and homomorphic encryption may affect real-time performance. Adaptive defenses and hybrid approaches such as blockchain-quantum technologies may improve its robustness and functionality in intelligent environments. Future research can improve the scalability, efficacy, and feasibility of Q-P2FL in smart environments. Furthermore, future work may focus on novel adversarial strategies, testing on a wide range of datasets and real-world settings, and optimizing computational efficiency.

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